Aror University of Art, Architecture, Design & Heritage Sukkur.

BS(Artificial Intelligence)
Fall-2023
Robotics

Course Title:	Robotics
Course Code:	
Credit Hours:	(2+1)
Course Instructor:	Bilal Ahmed
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Description:

This lab course provides hands-on experience corresponding to the theoretical concepts covered in the Introduction to Robotics lecture component. Students will work with microcontrollers (Arduino, ESP32), interface a range of sensors and actuators, build IoT-connected systems, and progress to robot modeling and simulation using ROS 2 and Gazebo. Each week's lab session directly reinforces that week's theory, building toward a final integrated robotics project that combines embedded systems, sensing, actuation, and ROS-based control.

Aims and Objectives:

1. Configure and program microcontroller platforms (Arduino, ESP32) to control digital and analog I/O, including LEDs, buzzers, and PWM-based outputs.
2. Interface and calibrate a variety of sensors (touch, analog, ultrasonic, IR) and actuators (DC, servo, stepper motors) to build functional sensor–actuator systems.
3. Implement IoT-based communication and ROS 2 fundamentals, including publisher–subscriber nodes, robot description (URDF), and coordinate frame transformations (TF).
4. Build and test a complete differential drive robot in both ROS 2 and Gazebo simulation, culminating in an integrated real-life robotics project.

Assessment:

S. No	**Assessment Activities**	Percentage	Total Activities
1.	Sessional: Quizzes/ Assignments (Quizzes & Assignments)	30%	Multiple
2.	Mid Term Exam	30%	1
3.	Final Exam	40%	1

Course Learning Outcomes (CLOs):

No.	Course Learning Outcome	Domain	GA	Assessment Tool
C1	**Understand** basic robotic problem-solving steps and control logic, including sensor-driven decision making and conditional/loop-based control constructs.	C2	GA2	Class Participation, Quizzes, Mid Exams., Assignments, Final Exam
C2	**Demonstrate** basic embedded programming concepts (Arduino/ESP32 in C++, ROS 2 in Python) for controlling sensors, actuators, and robot behavior.	C3	GA3	Class Activity, Quiz, Assignments, Mid Exam, Final Exam
C3	**Design and implement** algorithms to solve real-world robotics problems, such as obstacle avoidance, sensor data processing, and differential-drive robot navigation.	C3	GA4	Worksheets, Project

Domains:
C=Cognitive, A=Affective, P=Psychomotor0
Levels:
Cognitive = {1: Remembering, 2: Understanding, 3: Applying, 4: Analyzing, 5: Evaluating, 5: Creating}
Affective = {1: Receiving, 2: Responding, 3: Valuing, 4: Organizing, 5: Characterizing}
Psychomotor= {1: Imitation, 2: Manipulation, 3: Precision, 4: Articulation, 5: Naturalization}

Course Outlines:

Week	Lab Session — Hands-on Task	Completion %
1	Explore real-world robots (videos/demos); introduction to Arduino IDE; blink LED using Arduino	6%
2	Use Arduino digital and analog pins in read/write modes	13%
3	Read touch sensor values and control an LED; display messages/sensor values on Serial Monitor	19%
4	Read potentiometer or LDR values using Arduino; dim LED brightness using PWM	25%
5	Generate simple tones/alerts using buzzer	31%
6	Interface ultrasonic sensor with Arduino, measure distance, detect objects, and control LED	38%
7	Drive DC motor and rotate servo motor to specific angles	44%
8	Develop a small Arduino project prototype (sensor + actuator integration)	50%
9	Blink LED and control GPIO using ESP32; create a webpage to control LED/motor and monitor sensor values over Wi-Fi	56%
10	Send sensor data to cloud (ThingSpeak or Streamlit)	63%
11	Run publisher–subscriber examples and explore ROS message types	69%
12	Build and visualize a simple robot URDF (e.g. DDBot skeleton) in RViz; explore its coordinate frame tree	75%
13	Verify differential drive kinematics: compute expected v_L/v_R from given (v, ω), publish to robot, compare predicted vs. actual motion; drive DDBot via teleop	81%
14	Simulate built-in robots in Gazebo	88%
15	Develop and drive a differential drive robot in Gazebo	94%
16	Final project development and integration (real-life robotics project combining learned modules)	100%

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Text Books:

- Introduction to Robotics: Mechanics and Control by John J. Craig
- A Concise Introduction to Robot Programming with ROS2" by Francisco Martín Rico

Reference Books:

Books:

- A Concise Introduction to Robot Programming with ROS2
- Arduino Robotics" by John-David Warren, Josh Adams, and Harald Molle
- **"Robotics, Vision and Control: Fundamental Algorithms"** by Peter Corke

Prepared by____________________ Verified by__________________

HOD_____________________________